Block practice · Radians, sectors, triangles and 3D trigonometry

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Arc length, sector area and perimeter

Angles and areas of triangles inside a solid

Volume formulas for cones, pyramids and prisms using trig

Sector problems with an unknown radius or angle

Areas and perimeters of shapes made from sectors and triangles

Angle between two sloping faces with the cosine rule

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Practice workbook · Arc length, sector area and perimeter

Using s = r theta and A = (1/2) r^2 theta with theta in radians to find an arc, a sector area or a perimeter, or one unknown from a single given fact.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-03

1. Question 1 · part b · 2 marks

2022 JAN · Paper 02

Worked-example retry · Attempt part b only.

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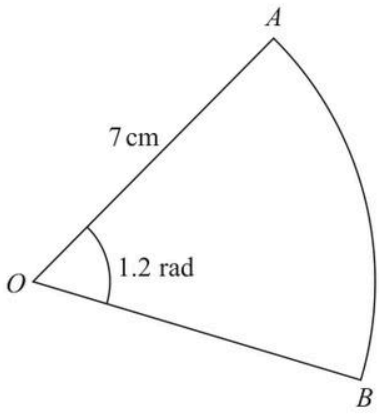


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows sector AOB of a circle with centre O and radius 7 cm.

The angle AOB is 1.2 radians.

Calculate

(a) the area of sector AOB **(2)**

(b) the perimeter of sector AOB **(2)**

(Total for Question 1 is 4 marks)

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1. Question 1 · part b · 2 marks · Working and answer

Working and answer

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2. Question 1 · part a · 1 mark

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

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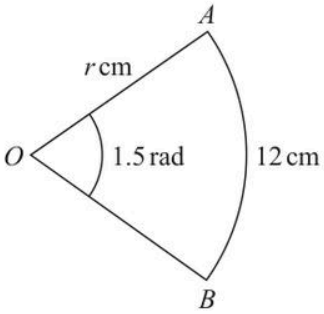


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows sector AOB of a circle with centre O and radius r cm. The angle AOB is 1.5 radians and the length of arc AB is 12 cm.

Calculate

(a) the value of r,

(1)

(b) the area of the sector AOB.

(2)

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(Total for Question 1 is 3 marks)

2. Question 1 · part a · 1 mark · Working and answer

Working and answer

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3. Question 1 · part b · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

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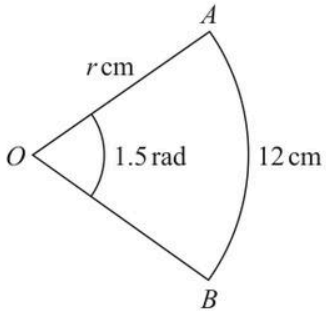


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Calculate

(a) the value of r,

(1)

(b) the area of the sector AOB.

(2)

(Total for Question 1 is 3 marks)

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3. Question 1 · part b · 2 marks · Working and answer

Working and answer

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4. Question 1 · part a · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part a only.

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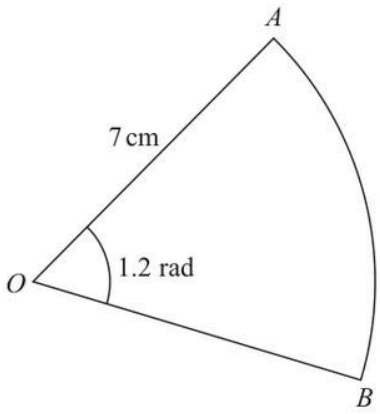


Diagram **NOT** accurately drawn

Figure 1

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The angle AOB is 1.2 radians.

Calculate

(a) the area of sector AOB **(2)**

(b) the perimeter of sector AOB **(2)**

(Total for Question 1 is 4 marks)

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4. Question 1 · part a · 2 marks · Working and answer

Working and answer

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5. Question 3 · part a · 2 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

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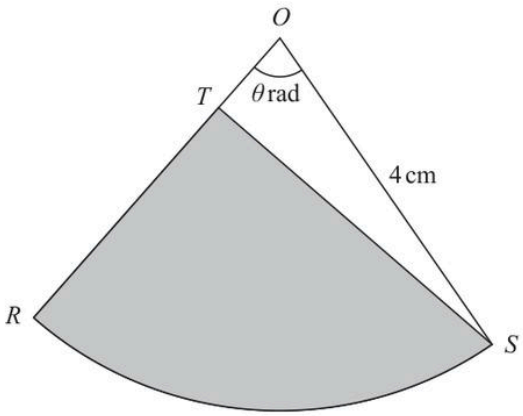


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows sector ORS of a circle with centre O and radius 4 cm.
The size of angle ROS is θ radians.

The area of sector ORS is 2π cm^2

(a) Find the exact value of θ

(2)

(b) Find the perimeter, in cm to 3 significant figures, of the sector ORS.

(2)

The point T lies on OR such that $OT : TR = 1 : 3$

The region shown shaded in Figure 1 is bounded by the line TR, the line TS and the arc RS of the sector.

The area of this region is A cm^2

(c) Find the exact value of A

(2)

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5. Question 3 · part a · 2 marks · Working and answer

Working and answer

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6. Question 3 · part b · 2 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

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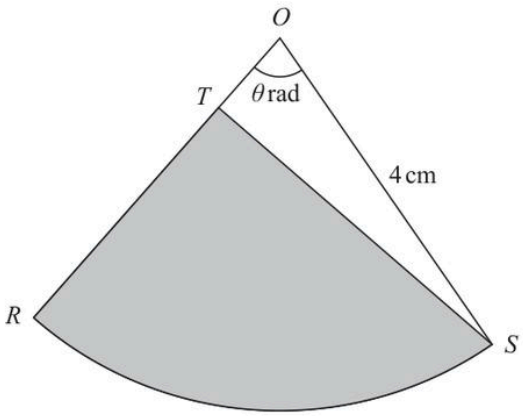


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(b) Find the perimeter, in cm to 3 significant figures, of the sector ORS.

(2)

The point T lies on OR such that $OT : TR = 1 : 3$

The region shown shaded in Figure 1 is bounded by the line TR, the line TS and the arc RS of the sector.

The area of this region is A cm^2

(c) Find the exact value of A

(2)

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6. Question 3 · part b · 2 marks · Working and answer

Working and answer

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7. Question 1 · part a · 1 mark

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

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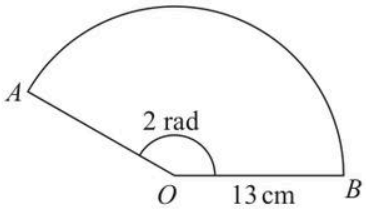


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the sector AOB of a circle with centre O.
The radius of the circle is 13 cm and angle $AOB = 2$ radians.

(a) Find the length of the arc AB.

(1)

(b) Find the area of the sector AOB.

(2)

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(Total for Question 1 is 3 marks)

7. Question 1 · part a · 1 mark · Working and answer

Working and answer

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8. Question 1 · part b · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

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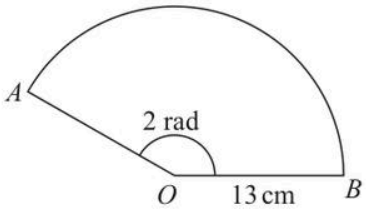


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the sector AOB of a circle with centre O.
The radius of the circle is 13 cm and angle $AOB = 2$ radians.

(a) Find the length of the arc AB.

(1)

(b) Find the area of the sector AOB.

(2)

(Total for Question 1 is 3 marks)

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8. Question 1 · part b · 2 marks · Working and answer

Working and answer

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9. Question 3 · part a · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

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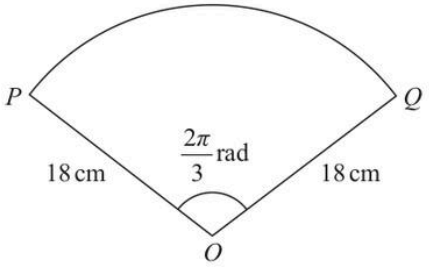


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a sector OPQ of a circle with centre O.

The radius of the circle is 18 cm and the angle POQ is $\frac{2\pi}{3}$ radians.

(a) Find the length of the arc PQ, giving your answer as a multiple of π

(2)

Figure 2 below shows the sector OPQ and the kite $OPTQ$.

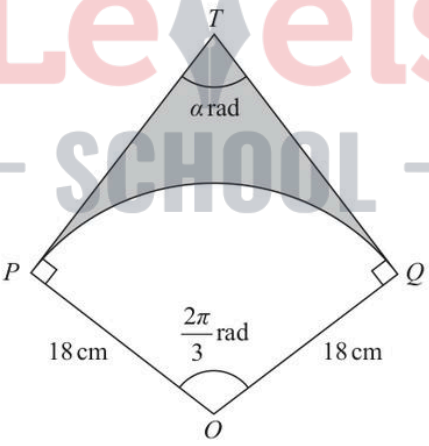


Diagram **NOT** accurately drawn

Figure 2

PT is the tangent to the circle at P and QT is the tangent at Q, such that angle $PTQ = \alpha$ radians.

(b) (i) Find α in terms of π

(1)

(ii) Calculate, to 3 significant figures, the area of the region, shown shaded in Figure 2, which is bounded by the arc PQ and the tangents PT and QT.

(6)

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9. Question 3 · part a · 2 marks · Working and answer

Working and answer

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10. Question 1 · part a · 2 marks

2024 NOV · Paper 02

Further practice · Attempt part a only.

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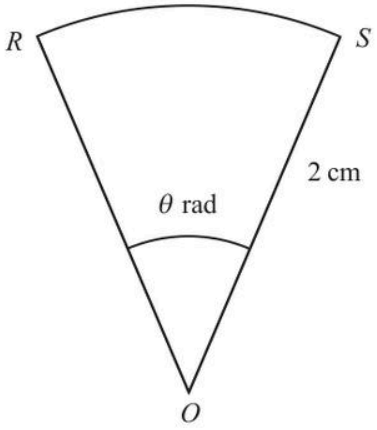


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows sector ROS of a circle with centre O and radius 2 cm
The size of angle ROS is θ radians.

The area of sector ROS is $\frac{\pi}{2}$ cm^2

(a) Find the exact value of θ

(2)

The perimeter of sector ROS is P cm

(b) Find the exact value of P

(3)

(Total for Question 1 is 5 marks)

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10. Question 1 · part a · 2 marks · Working and answer

Working and answer

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11. Question 1 · part b · 3 marks

2024 NOV · Paper 02

Further practice · Attempt part b only.

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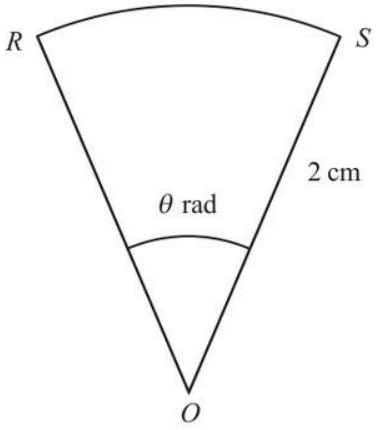


Diagram **NOT** accurately drawn

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The size of angle ROS is θ radians.

The area of sector ROS is $\frac{\pi}{2}$ cm^2

(a) Find the exact value of θ

(2)

The perimeter of sector ROS is P cm

(b) Find the exact value of P

(3)

(Total for Question 1 is 5 marks)

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11. Question 1 · part b · 3 marks · Working and answer

Working and answer

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Practice workbook · Angles and areas of triangles inside a solid

Finding an angle or area of a triangle drawn inside a solid (a face or a slanted cross-section) by first finding its sides and then using trig or the cosine rule.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-18

1. Question 6 · part b · 2 marks

2023 NOV · Paper 01

Worked-example retry · Attempt part b only.

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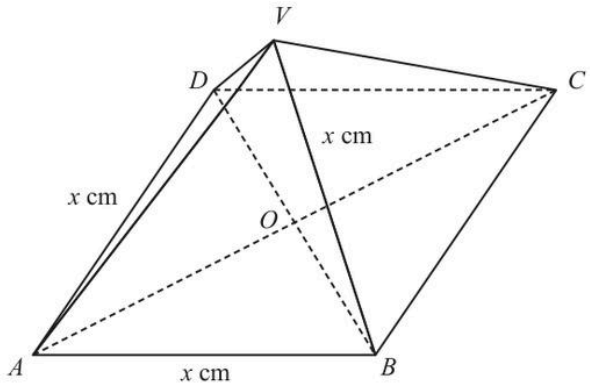


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right pyramid with a horizontal square base.

$$AB = BC = CD = DA = x \text{ cm}$$

$$AV = BV = CV = DV = x \text{ cm}$$

O is the point of intersection of the diagonals of the base.
The vertex V of the pyramid is vertically above O

(a) Show that $VO = \frac{\sqrt{2}}{2}x$ cm

(3)

(b) Find, in degrees, the size of the angle AVC

(2)

(c) Find, in degrees to one decimal place, the size of the angle between the plane VAB and the plane VDC

(3)

The volume of the pyramid is 200 cm^3

Given that the volume of a pyramid $= \frac{1}{3} \times$ base area $\times$ height

(d) Find to 3 signficant figures, the value of x

(3)

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1. Question 6 · part b · 2 marks · Working and answer

Working and answer

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2. Question 8 · part d · 6 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

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Diagram **NOT** accurately drawn

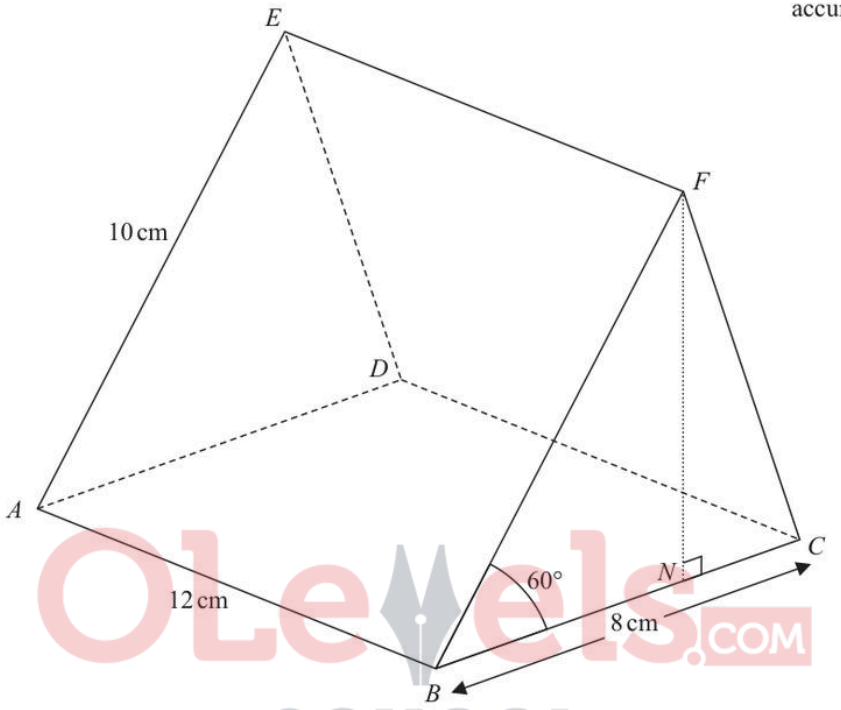


Figure 3

Figure 3 shows a right prism $ABCDEF$. The cross section BCF of the prism is a triangle.

$AB = DC = 12\,\text{cm}$ $BC = AD = 8\,\text{cm}$ $BF = AE = 10\,\text{cm}$ $\angle FBC = \angle EAD = 60°$

The point N lies on BC such that FN is perpendicular to BC.

(a) Show that $BN = 5\,\text{cm}$.

(2)

(b) Find, in cm to 3 significant figures, the length of EN.

(3)

The midpoint of BF is X and the midpoint of FC is Y.

(c) Find, in degrees to one decimal place, the size of the angle between the plane $ABCD$ and the plane $AXYD$.

(2)

(d) Find, in degrees to one decimal place, the size of the angle AYE.

(6)

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P 6 1 8 8 3 A 0 2 2 3 6

2. Question 8 · part d · 6 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 8 continued

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2. Question 8 · part d · 6 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 8 continued

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2. Question 8 · part d · 6 marks

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Further practice · Attempt part d only.

Question 8 continued

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(Total for Question 8 is 13 marks)

2. Question 8 · part d · 6 marks · Working and answer

Working and answer

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3. Question 4 · part b · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part b only.

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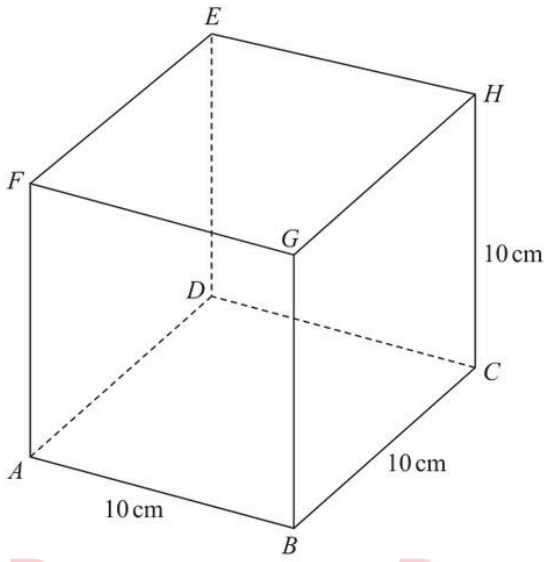


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a solid cube *ABCDEFGH* with sides of length 10 cm.

(a) Find, in cm, the exact length of *FC*

(2)

(b) Find, in degrees to one decimal place, the size of $\angle FCA$

(2)

(c) Find, to the nearest degree, the size of the obtuse angle between the plane *BFH* and the plane *BHC*

(5)

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P 6 6 3 0 8 A 0 8 3 6

3. Question 4 · part b · 2 marks · Working and answer

Working and answer

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4. Question 10 · part c · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

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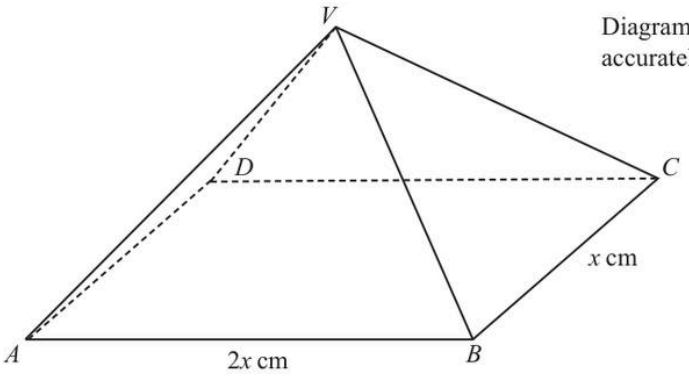


Figure 4

Figure 4 shows a right pyramid $ABCDV$

The base of the pyramid is a rectangle where,

$AB = DC = 2x$ cm $AD = BC = x$ cm

The edges VA, VB, VC and VD are all of equal length.

The angle between VA and $ABCD$ is 45°

(a) Show that $VA = \frac{\sqrt{10}}{2}x$ cm

(3)

(b) Find in cm, the exact height of the pyramid in terms of x

(2)

Find, in degrees to one decimal place,

(c) the size of angle VBA

(2)

(d) the size of the obtuse angle between the plane AVC and the plane BVD

(4)

Given that the volume of the pyramid is $9\sqrt{5}$ cm^3

(e) find the value of x

(2)

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4. Question 10 · part c · 2 marks · Working and answer

Working and answer

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5. Question 4 · part b · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

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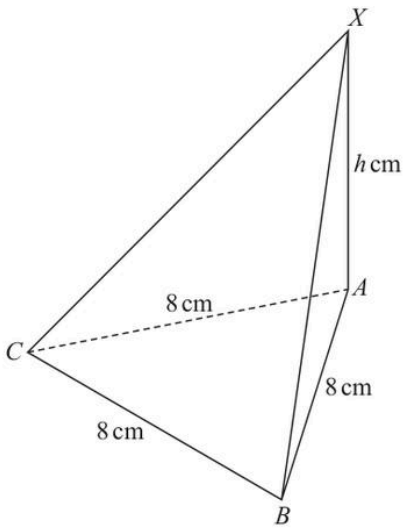


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a triangular pyramid $ABCX$.
The base ABC of the pyramid is an equilateral triangle where $AB = BC = CA = 8$ cm.
The vertex X of the pyramid is such that AX is perpendicular to the base of the pyramid and $AX = h$ cm.

The volume of the pyramid is $48\sqrt{3}$ cm^3

(a) Show that $h = 9$

(3)

(b) Find, in degrees to one decimal place, the size of angle BXC.

(3)

(c) Find, in degrees to one decimal place, the size of the angle between the plane BCX and the base ABC of the pyramid.

(3)

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5. Question 4 · part b · 3 marks · Working and answer

Working and answer

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6. Question 3 · part b · 3 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

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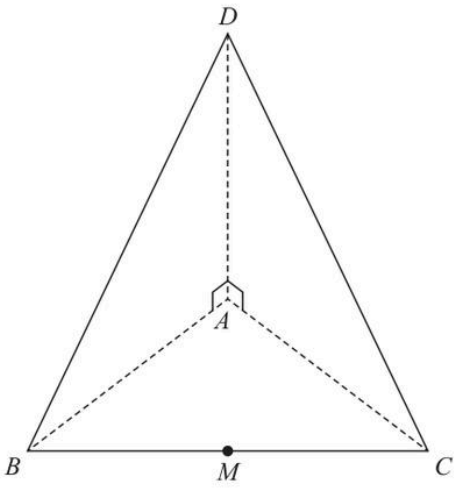


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a triangular pyramid $ABCD$.

The base, ABC, of the pyramid is a horizontal isosceles triangle with $AB = AC = 10\,\text{cm}$ and $BC = 16\,\text{cm}$. The midpoint of BC is M.

The face BCD of the pyramid is an isosceles triangle with $BD = CD = 26\,\text{cm}$ and D is vertically above A.

$\angle BAD = \angle CAD = 90°$

(a) Calculate the length, in cm, of AM.

(2)

Calculate, in degrees to the nearest degree,

(b) the size of $\angle BCD$,

(3)

(c) the size of the angle between the planes BCA and BCD.

(4)

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6. Question 3 · part b · 3 marks · Working and answer

Working and answer

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7. Question 8 · part c · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

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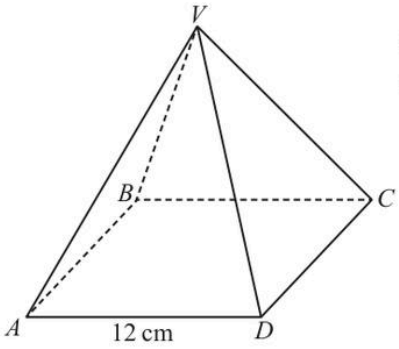


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right pyramid with vertex V and square base, $ABCD$, of side 12 cm

The size of the angle between the plane VAB and the base $ABCD$ is $30°$

(a) Show that the height of the pyramid is $2\sqrt{3}$ cm

(2)

(b) Find, in cm, the exact length of VA

(3)

(c) Find, in cm^2, the exact area of triangle VAD

(3)

(d) Find, in cm, the exact length of the perpendicular from D to VA

(2)

(e) Find, in degrees to one decimal place, the size of the obtuse angle between the plane VAB and the plane VAD

(3)

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7. Question 8 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 9 · part c · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

9

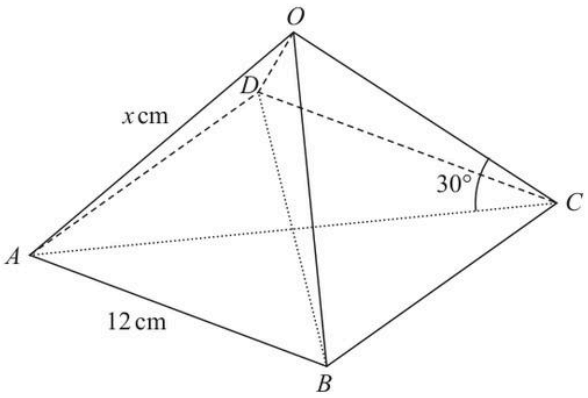


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows the right pyramid $OABCD$ with a square base $ABCD$ of side 12 cm.

$OA = OB = OC = OD = x$ cm and $\angle OAC = \angle ODB = \angle OCA = \angle OBD = 30°$

(a) Find the exact length of AC

(2)

(b) Show that $x = 4\sqrt{6}$

(2)

(c) Find the total surface area, to the nearest cm^2, of the pyramid.

(5)

(d) Find the size of the obtuse angle, to the nearest degree, between the plane OAB and the plane OBC

(4)

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P 7 1 8 1 8 A 0 2 2 2 8

8. Question 9 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Volume formulas for cones, pyramids and prisms using trig

Writing the volume of a cone, pyramid or prism in terms of one variable by using trig to link its dimensions, then solving or rearranging.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-TRI-19

1. Question 9 · part a · 5 marks

2025 NOV · Paper 01

Worked-example retry · Attempt part a only.

9

Diagram **NOT** accurately drawn

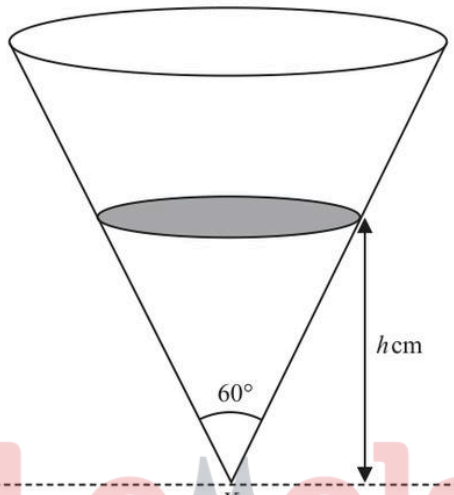


Figure 2

Figure 2 shows a hollow right circular cone fixed with its axis of symmetry vertical.

The vertical angle of the cone is 60°
Initially the cone is empty.

At time $t = 0$ liquid starts to fill the cone at a constant rate of $0.2\,\text{cm}^3/\text{s}$
At time t seconds after the liquid starts to fill the cone, the height of the liquid is h cm above X

(a) Show that $h = \sqrt[3]{\frac{9t}{5\pi}}$

(5)

The surface area of the liquid, shown shaded in Figure 2, is increasing at a constant rate of $p\,\text{cm}^2/\text{s}$ when $t = 6$

(b) Find, to 3 significant figures, the value of p

(8)

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OLevels.com SCHOOL

1. Question 9 · part a · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 10 · part e · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

10

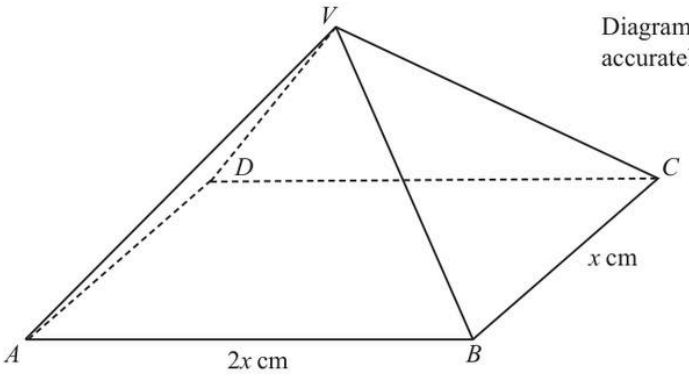


Figure 4

Figure 4 shows a right pyramid $ABCDV$

The base of the pyramid is a rectangle where,

$AB = DC = 2x$ cm $AD = BC = x$ cm

The edges VA, VB, VC and VD are all of equal length.

The angle between VA and $ABCD$ is 45°

(a) Show that $VA = \frac{\sqrt{10}}{2}x$ cm

(3)

(b) Find in cm, the exact height of the pyramid in terms of x

(2)

Find, in degrees to one decimal place,

(c) the size of angle VBA

(2)

(d) the size of the obtuse angle between the plane AVC and the plane BVD

(4)

Given that the volume of the pyramid is $9\sqrt{5}$ cm^3

(e) find the value of x

(2)

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26

P 7 4 2 8 5 R A 0 2 6 3 2

2. Question 10 · part e · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

Question 10 continued

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2. Question 10 · part e · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

Question 10 continued

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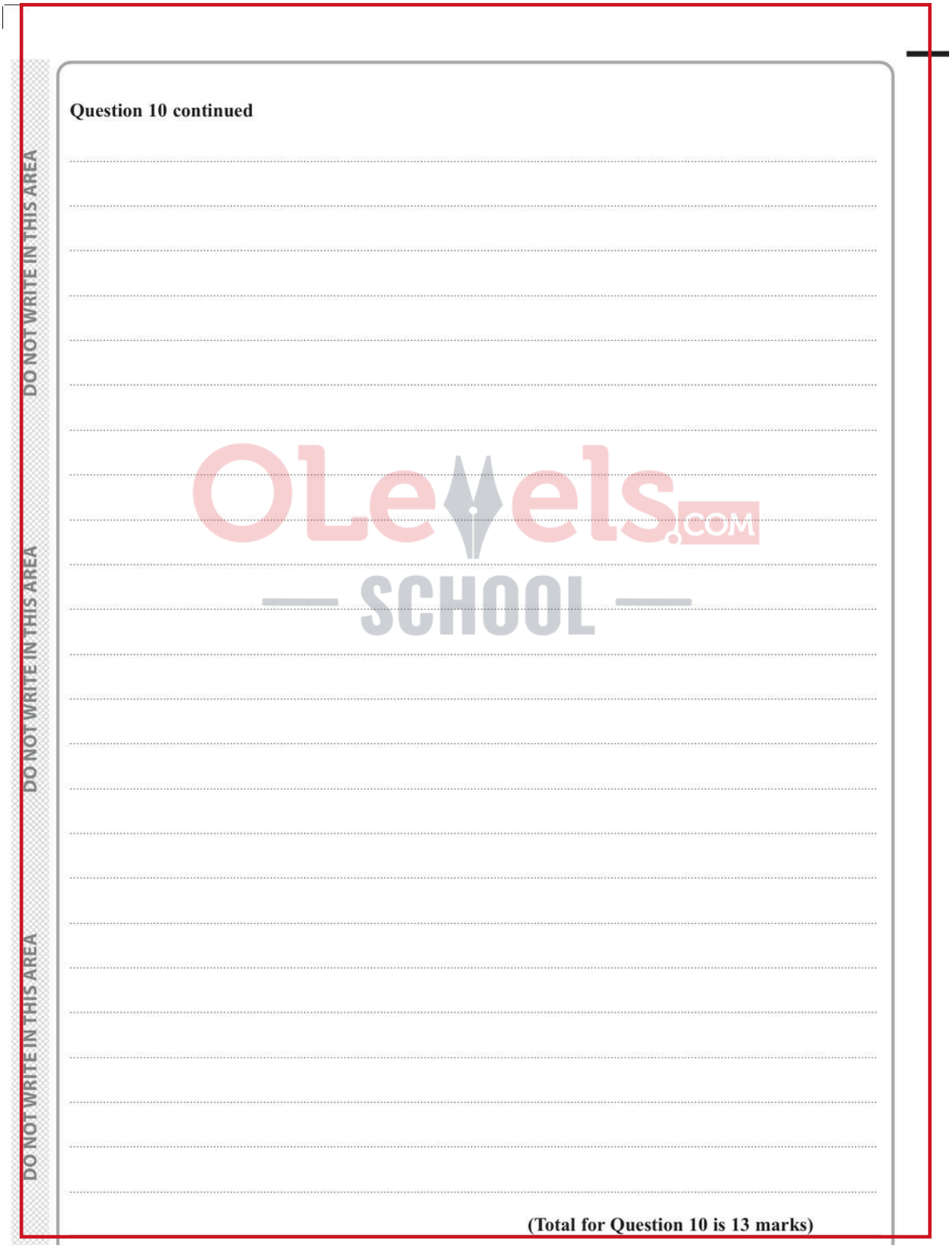
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OLevels.com SCHOOL

2. Question 10 · part e · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.



Question 10 continued

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OLevels.com SCHOOL

(Total for Question 10 is 13 marks)

2. Question 10 · part e · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 6 · part d · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part d only.

6

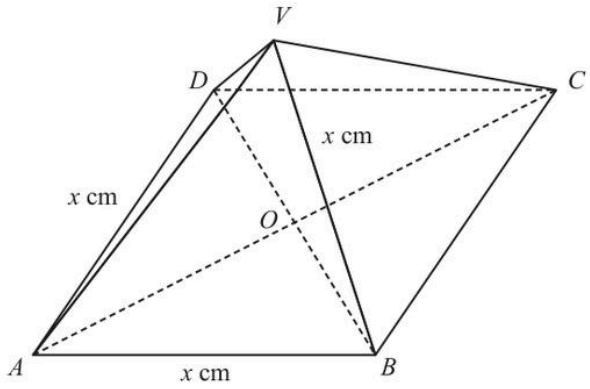


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right pyramid with a horizontal square base.

$$AB = BC = CD = DA = x \text{ cm}$$

$$AV = BV = CV = DV = x \text{ cm}$$

O is the point of intersection of the diagonals of the base.
The vertex V of the pyramid is vertically above O

(a) Show that $VO = \frac{\sqrt{2}}{2}x$ cm

(3)

(b) Find, in degrees, the size of the angle AVC

(2)

(c) Find, in degrees to one decimal place, the size of the angle between the plane VAB and the plane VDC

(3)

The volume of the pyramid is 200 cm^3

Given that the volume of a pyramid $= \frac{1}{3} \times$ base area $\times$ height

(d) Find to 3 signficant figures, the value of x

(3)

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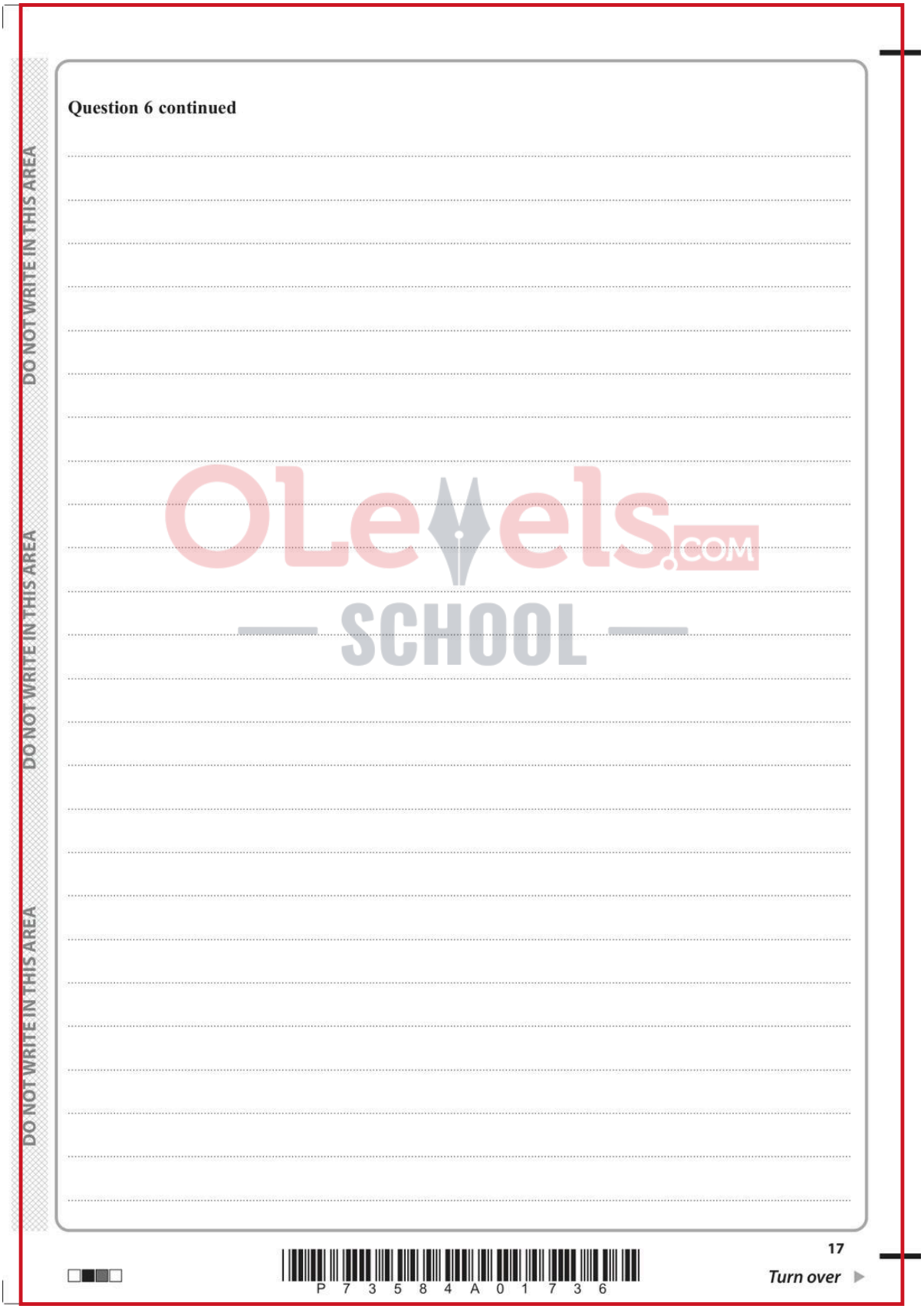
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3. Question 6 · part d · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part d only.



Question 6 continued

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OLevels.com SCHOOL

P 7 3 5 8 4 A 0 1 7 3 6

17

Turn over ▶

3. Question 6 · part d · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part d only.

Question 6 continued

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OLevels.com SCHOOL

3. Question 6 · part d · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part d only.

Question 6 continued

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(Total for Question 6 is 11 marks)

3. Question 6 · part d · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 4 · part a · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

4

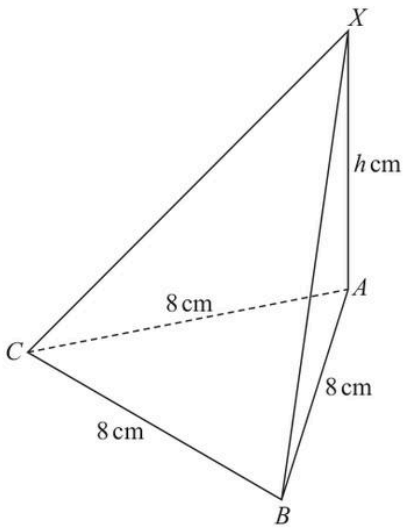


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a triangular pyramid $ABCX$.
The base ABC of the pyramid is an equilateral triangle where $AB = BC = CA = 8$ cm.
The vertex X of the pyramid is such that AX is perpendicular to the base of the pyramid and $AX = h$ cm.

The volume of the pyramid is $48\sqrt{3}$ cm^3

(a) Show that $h = 9$

(3)

(b) Find, in degrees to one decimal place, the size of angle BXC.

(3)

(c) Find, in degrees to one decimal place, the size of the angle between the plane BCX and the base ABC of the pyramid.

(3)

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8

P 6 0 4 7 6 A 0 8 4 0

4. Question 4 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 6 · part a · 3 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

6

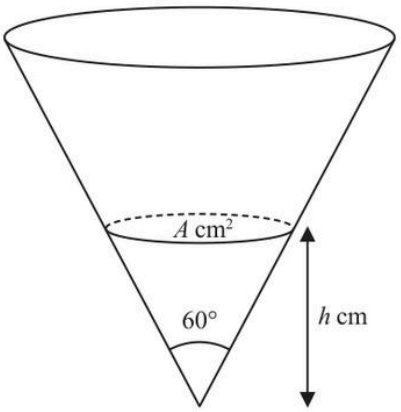


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a container in the shape of a right circular cone.
The container is fixed with its axis of symmetry vertical.
The vertical angle of the container is 60° as shown in the diagram.

At time t seconds, $t > 0$, the height of oil in the container is h cm and the volume of oil in the container is V cm^3

(a) Show that $V = \frac{1}{9}\pi h^3$

(3)

At time t seconds the surface area of oil in the container is A cm^2, as shown in Figure 4

Oil is dripping out of the bottom of the container at a constant rate of 4 cm^3/s.

(b) Find the exact rate of change, in cm^2/s, of the surface area of oil in the container when $h = 24$

(8)

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P 7 4 0 9 8 A 0 1 6 3 2

5. Question 6 · part a · 3 marks · Working and answer

Working and answer

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6. Question 6 · part a · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

6

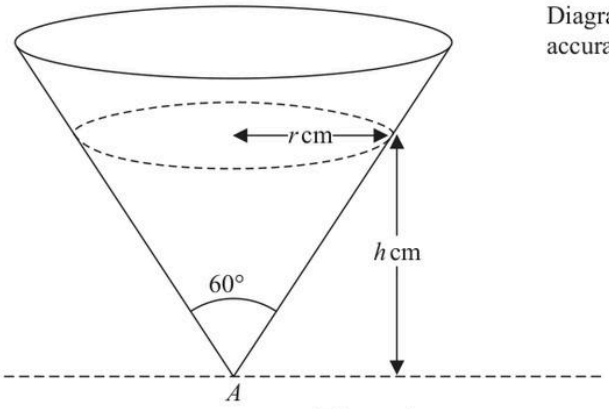


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a hollow right circular cone fixed with its axis of symmetry vertical.

The cone is inverted and contains liquid, which is dripping out of a small hole at the vertex A of the cone at a constant rate of $0.9\,\text{cm}^3/\text{s}$.

At time t seconds after the liquid starts to drip from the cone, the height of the liquid is h cm above A. The volume of liquid in the cone at time t seconds is $V\,\text{cm}^3$

The vertical angle of the cone is $60°$

(a) Show that $V = \frac{1}{9}\pi h^3$ **(2)**

(b) Find, in cm/s to 3 significant figures, the rate at which the height of the liquid is decreasing when the height of the liquid in the cone above the vertex is 1.2 cm. **(4)**

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P 6 0 4 7 6 A 0 1 6 4 0

6. Question 6 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 6 · part a · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part a only.

6

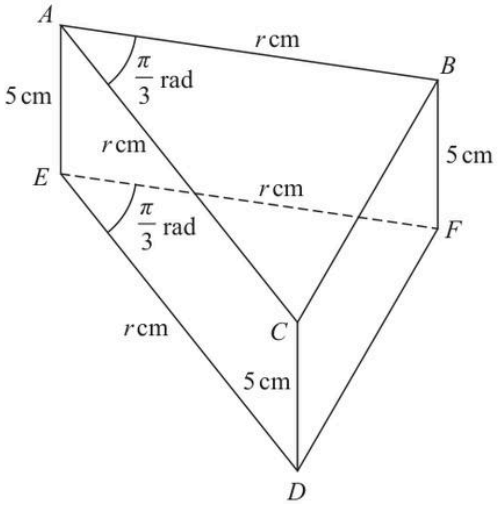


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right triangular prism $ABCDEF$. A cross section ABC of the prism is a triangle in which $AB = AC = r\,\text{cm}$ and $\angle CAB = \frac{\pi}{3}$ radians.

In the prism

$$AE = BF = CD = 5\,\text{cm} \qquad ED = EF = r\,\text{cm} \text{ and } \angle DEF = \frac{\pi}{3} \text{ radians}$$

(a) Show that the volume of the prism is $\frac{5\sqrt{3}}{4} r^2\ \text{cm}^3$

(1)

The volume of the prism is increasing in such a way that the size of $\angle CAB$ and the size of $\angle DEF$ remain constant and the length of AE, the length of BF and the length of CD remain constant.
The lengths of AB, AC, ED and EF are each increasing at a constant rate of $0.2\,\text{cm/s}$

(b) Find the exact rate of increase, in cm^3/s, of the volume of the prism when the area of the rectangular face $BCDF$ is $60\,\text{cm}^2$

(5)

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P 7 2 8 6 6 A 0 1 6 3 6

7. Question 6 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Sector problems with an unknown radius or angle

Setting up and solving equations in r and theta from given facts about a sector, such as its area and perimeter together, or a sector rolled into a cone.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-04

1. Question 3 · part whole · 8 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part whole only.

3

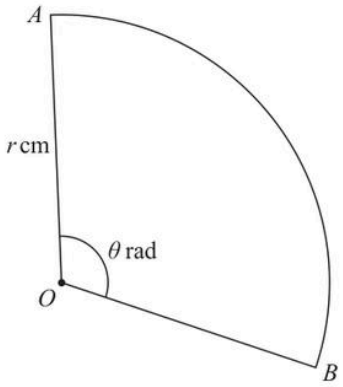


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the sector AOB of a circle with centre O and radius r cm, where r is an integer. The size of angle AOB is θ radians.

The sector has an area of 16.8 cm^2 and a perimeter of 16.4 cm.

Calculate

(i) the value of r

(ii) the value of θ

(8)

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1. Question 3 · part whole · 8 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part whole only.

Question 3 continued

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(Total for Question 3 is 8 marks)

1. Question 3 · part whole · 8 marks · Working and answer

Working and answer

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2. Question 3 · part whole · 6 marks

2023 JUN · Paper 02

Further practice · Attempt part whole only.

3

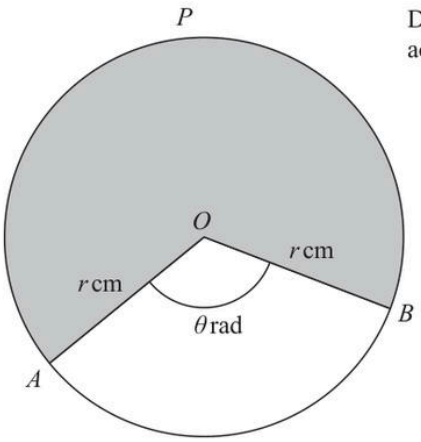


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a circle, centre O, with radius r cm.

The points A, P and B lie on the circle.

The obtuse angle $AOB = \theta$ radians.

The area of the sector $APBO$, shown shaded, is $372.4\,\text{cm}^2$ and the length of the arc APB is 53.2 cm.

Find, to 3 significant figures where appropriate, the value of

(i) r

(ii) θ

(6)

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P 7 2 8 6 6 A 0 6 3 6

2. Question 3 · part whole · 6 marks

2023 JUN · Paper 02

Further practice · Attempt part whole only.

Question 3 continued

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(Total for Question 3 is 6 marks)

2. Question 3 · part whole · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 4 · part a · 3 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

4

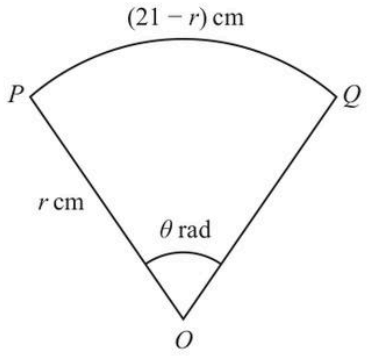


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows the sector OPQ of a circle with centre O and radius r cm.

$OP = OQ = r$ cm arc $PQ = (21 - r)$ cm $\angle POQ = \theta$ radians

The area of the sector is A cm^2

(a) Show that $A = \frac{r}{2}(21 - r)$ **(3)**

The area of the sector must be greater than or equal to 27 cm^2

(b) Find the set of possible values of r **(4)**

(c) Hence write down the set of possible values of θ **(2)**

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P 7 4 0 9 9 A 0 8 3 2

3. Question 4 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 4 · part c · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

4

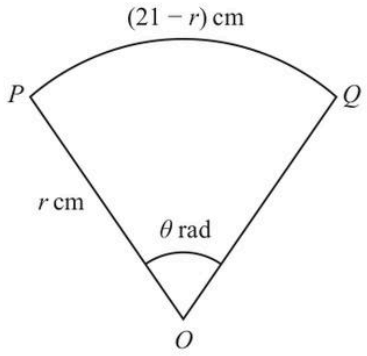


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows the sector OPQ of a circle with centre O and radius r cm.

$OP = OQ = r$ cm arc $PQ = (21 - r)$ cm $\angle POQ = \theta$ radians

The area of the sector is A cm^2

(a) Show that $A = \frac{r}{2}(21 - r)$ **(3)**

The area of the sector must be greater than or equal to 27 cm^2

(b) Find the set of possible values of r **(4)**

(c) Hence write down the set of possible values of θ **(2)**

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4. Question 4 · part c · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 9 marks)

4. Question 4 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 1 · part whole · 3 marks

2025 NOV · Paper 02

Further practice · Attempt part whole only.

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1 A circle has centre O and radius 14 cm

A sector AOB of the circle has a perimeter of 49 cm

Find, in cm^2, the area of sector AOB

(3)

OLevels.com — SCHOOL —

(Total for Question 1 is 3 marks)

5. Question 1 · part whole · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 3 · part a · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

3

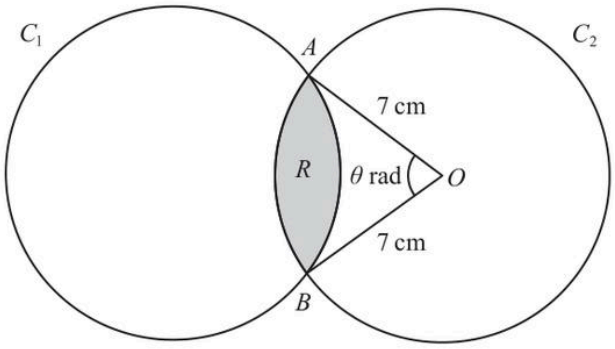


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows circle C_1 with radius 7 cm and circle C_2 with centre O and radius 7 cm

The circles C_1 and C_2 intersect at the point A and at the point B

The size of angle AOB is θ radians

The perimeter of the region R shown shaded in Figure 1 is $\frac{28\pi}{9}$ cm

(a) Find the exact value of θ **(2)**

(b) Find the area, in cm^2 to 3 significant figures, of the region R **(4)**

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P 7 2 8 6 5 A 0 1 0 3 6

6. Question 3 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 10 · part a · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

10

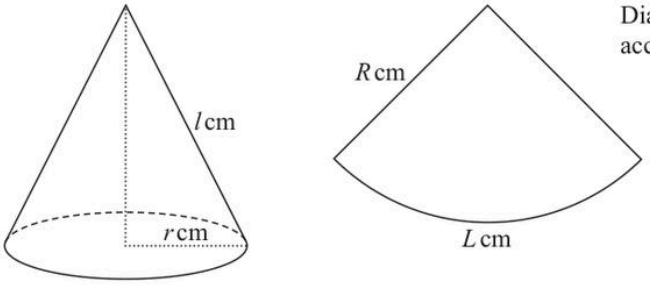


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a right circular cone with base radius r cm and slant height l cm.
Figure 4 also shows a sector of a circle with radius R cm and arc length L cm.

The area of the curved surface of the cone is A cm^2

By considering how the sector of the circle can be folded to exactly form the curved surface of the cone with R and L suitably chosen,

(a) prove that $A = \pi rl$

(4)

Sand is poured onto a horizontal surface at a constant rate of 1.5 cm^3/s.
The sand forms a pile in the shape of a right circular cone with its base on the surface.
The curved surface area of the cone, A cm^2, increases in such a way that the height of the cone is always three times the radius of the base of the cone.

Given that $\frac{dA}{dr} = k\pi r$, where k is a constant,

(b) find the exact value of k.

(3)

(c) Calculate the rate, in cm^2/s, to 3 significant figures, at which the curved surface area of the pile is increasing when the height of the pile is 24 cm.

(5)

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P 6 2 2 8 0 A 0 2 6 3 2

7. Question 10 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Areas and perimeters of shapes made from sectors and triangles

Finding the area or perimeter of a region built from sectors, segments, semicircles and triangles, using (1/2)ab sin C and the cosine rule for straight edges.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-05

1. Question 4 · part whole · 7 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part whole only.

4

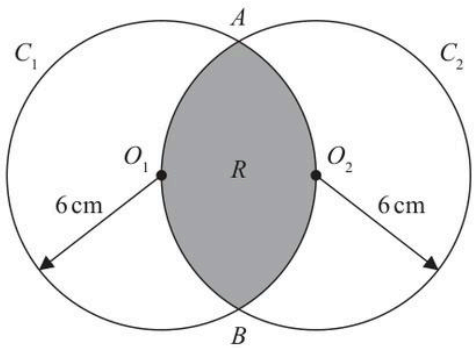


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows two circles, C_1 and C_2, each with a radius of 6 cm.

The centre of C_1 is O_1 such that O_1 lies on C_2

The centre of C_2 is O_2 such that O_2 lies on C_1

The circles intersect at the points A and B and enclose the region R, shown shaded in Figure 1

The area of region R is P cm^2

Find the exact value of P, giving your answer in the form $a\pi - b\sqrt{c}$ where a, b and c are integers.

(7)

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8

P 7 1 6 6 6 A 0 8 3 6

1. Question 4 · part whole · 7 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part whole only.

Question 4 continued

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1. Question 4 · part whole · 7 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part whole only.

Question 4 continued

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1. Question 4 · part whole · 7 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part whole only.

Question 4 continued

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(Total for Question 4 is 7 marks)

1. Question 4 · part whole · 7 marks · Working and answer

Working and answer

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2. Question 3 · part c · 2 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

3

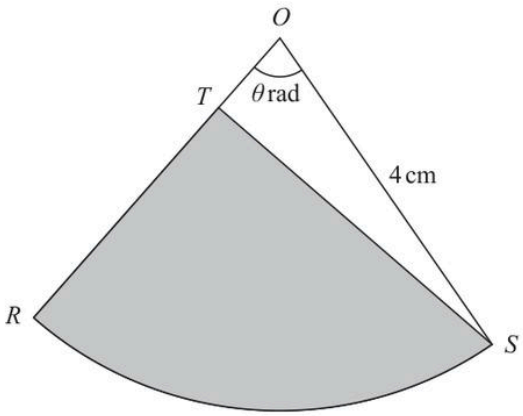


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows sector ORS of a circle with centre O and radius 4 cm.
The size of angle ROS is θ radians.

The area of sector ORS is 2π cm^2

(a) Find the exact value of θ

(2)

(b) Find the perimeter, in cm to 3 significant figures, of the sector ORS.

(2)

The point T lies on OR such that $OT : TR = 1 : 3$

The region shown shaded in Figure 1 is bounded by the line TR, the line TS and the arc RS of the sector.

The area of this region is A cm^2

(c) Find the exact value of A

(2)

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6

P 7 1 6 4 1 A 0 6 3 6

2. Question 3 · part c · 2 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 3 continued

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(Total for Question 3 is 6 marks)

2. Question 3 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 9 · part a · 7 marks

2023 NOV · Paper 02

Further practice · Attempt part a only.

9

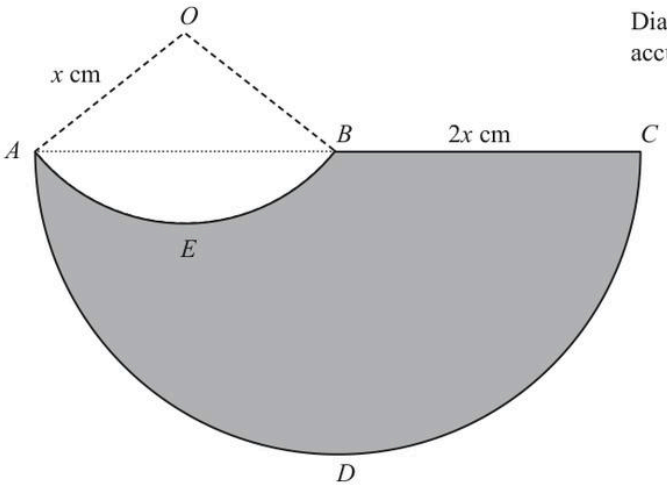


Diagram **NOT** accurately drawn

Figure 2

A logo, $AEBCD$, is shown shaded in Figure 2.

The straight line ABC is the diameter of the semicircle ADC
AEB is an arc of a circle with centre O
All angles are measured in radians.

- $BC = 2x$ cm
- $OA = OB = x$ cm
- length of arc $AEB = 1.8x$ cm

The perimeter of the logo is P

(a) Show that $P = ax(\pi + \pi \sin 0.9 + b)$ where a and b are constants to be found.

(7)

Given that $x = 10$ cm,

(b) find, in cm^2 to 3 significant figures, the area of the logo.

(6)

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3. Question 9 · part a · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 9 · part b · 6 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

9

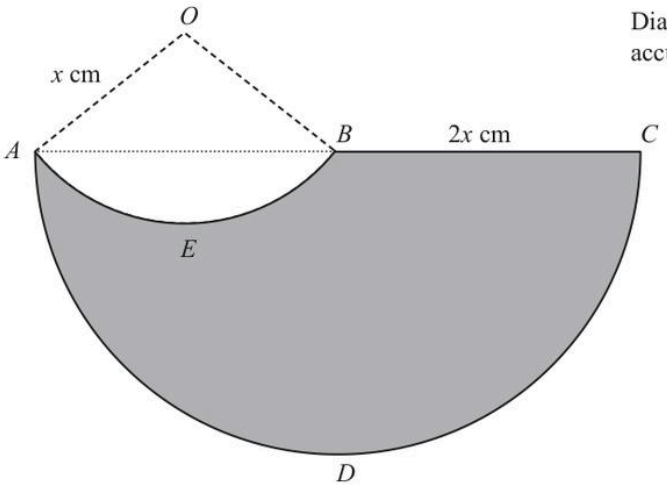


Diagram **NOT** accurately drawn

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(7)

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(6)

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4. Question 9 · part b · 6 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

Question 9 continued

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4. Question 9 · part b · 6 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

Question 9 continued

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4. Question 9 · part b · 6 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

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Question 9 continued

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(Total for Question 9 is 13 marks)

4. Question 9 · part b · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 3 · part b · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

3

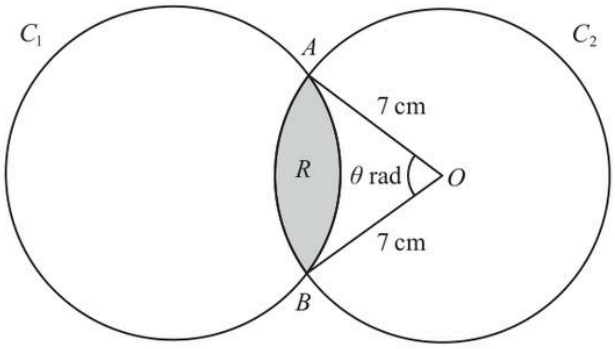


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows circle C_1 with radius 7 cm and circle C_2 with centre O and radius 7 cm

The circles C_1 and C_2 intersect at the point A and at the point B

The size of angle AOB is θ radians

The perimeter of the region R shown shaded in Figure 1 is $\frac{28\pi}{9}$ cm

(a) Find the exact value of θ **(2)**

(b) Find the area, in cm^2 to 3 significant figures, of the region R **(4)**

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5. Question 3 · part b · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 6 marks)

5. Question 3 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 3 · part whole · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

3

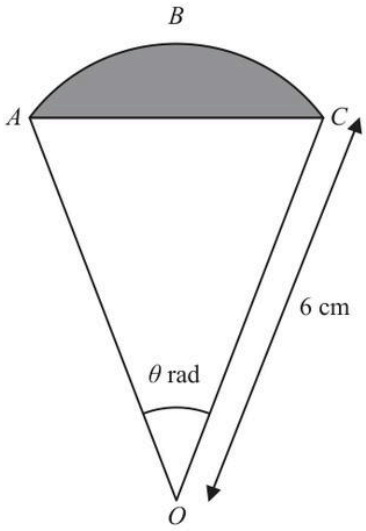


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows sector $OABC$ of a circle with centre O and radius 6 cm
The size of angle AOC is θ radians.

The perimeter of sector $OABC$ is $(12+\pi)$ cm

The region shown shaded in Figure 1 is bounded by the arc ABC and the line AC
The area of this region is S cm^2

Find the exact value of S

(6)

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6. Question 3 · part whole · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 3 continued

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(Total for Question 3 is 6 marks)

6. Question 3 · part whole · 6 marks · Working and answer

Working and answer

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7. Question 4 · part a · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

4

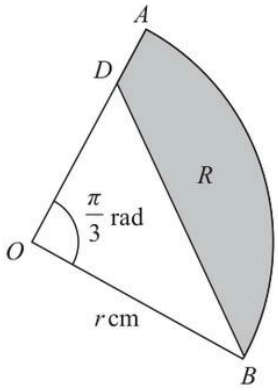


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows sector OAB of a circle with centre O and radius r cm.

The angle $AOB = \frac{\pi}{3}$ radians.

The point D divides OA in the ratio $3 : 1$

The area of the region R, shown shaded in Figure 1, is $79.5\,\text{cm}^2$

(a) Calculate the value, to 2 significant figures, of r

(4)

(b) Calculate the perimeter, in cm to 2 significant figures, of the region R

(4)

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7. Question 4 · part a · 4 marks · Working and answer

Working and answer

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8. Question 4 · part b · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

4

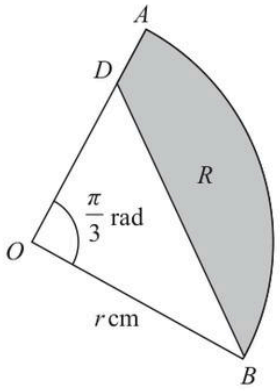


Diagram **NOT** accurately drawn

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The area of the region R, shown shaded in Figure 1, is $79.5\,\text{cm}^2$

(a) Calculate the value, to 2 significant figures, of r

(4)

(b) Calculate the perimeter, in cm to 2 significant figures, of the region R

(4)

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8. Question 4 · part b · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 8 marks)

8. Question 4 · part b · 4 marks · Working and answer

Working and answer

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Practice workbook · Angle between two sloping faces with the cosine rule

Finding the angle between two slanted faces of a solid by dropping perpendiculars from each face onto their common edge and using the cosine rule.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-17

1. Question 8 · part e · 3 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part e only.

8

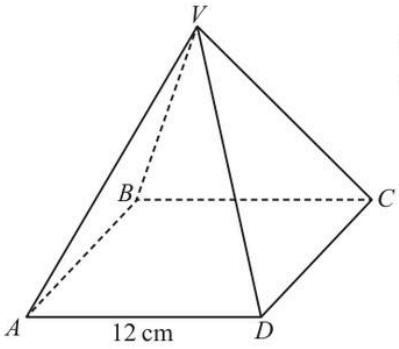


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right pyramid with vertex V and square base, $ABCD$, of side 12 cm

The size of the angle between the plane VAB and the base $ABCD$ is $30°$

(a) Show that the height of the pyramid is $2\sqrt{3}$ cm

(2)

(b) Find, in cm, the exact length of VA

(3)

(c) Find, in cm^2, the exact area of triangle VAD

(3)

(d) Find, in cm, the exact length of the perpendicular from D to VA

(2)

(e) Find, in degrees to one decimal place, the size of the obtuse angle between the plane VAB and the plane VAD

(3)

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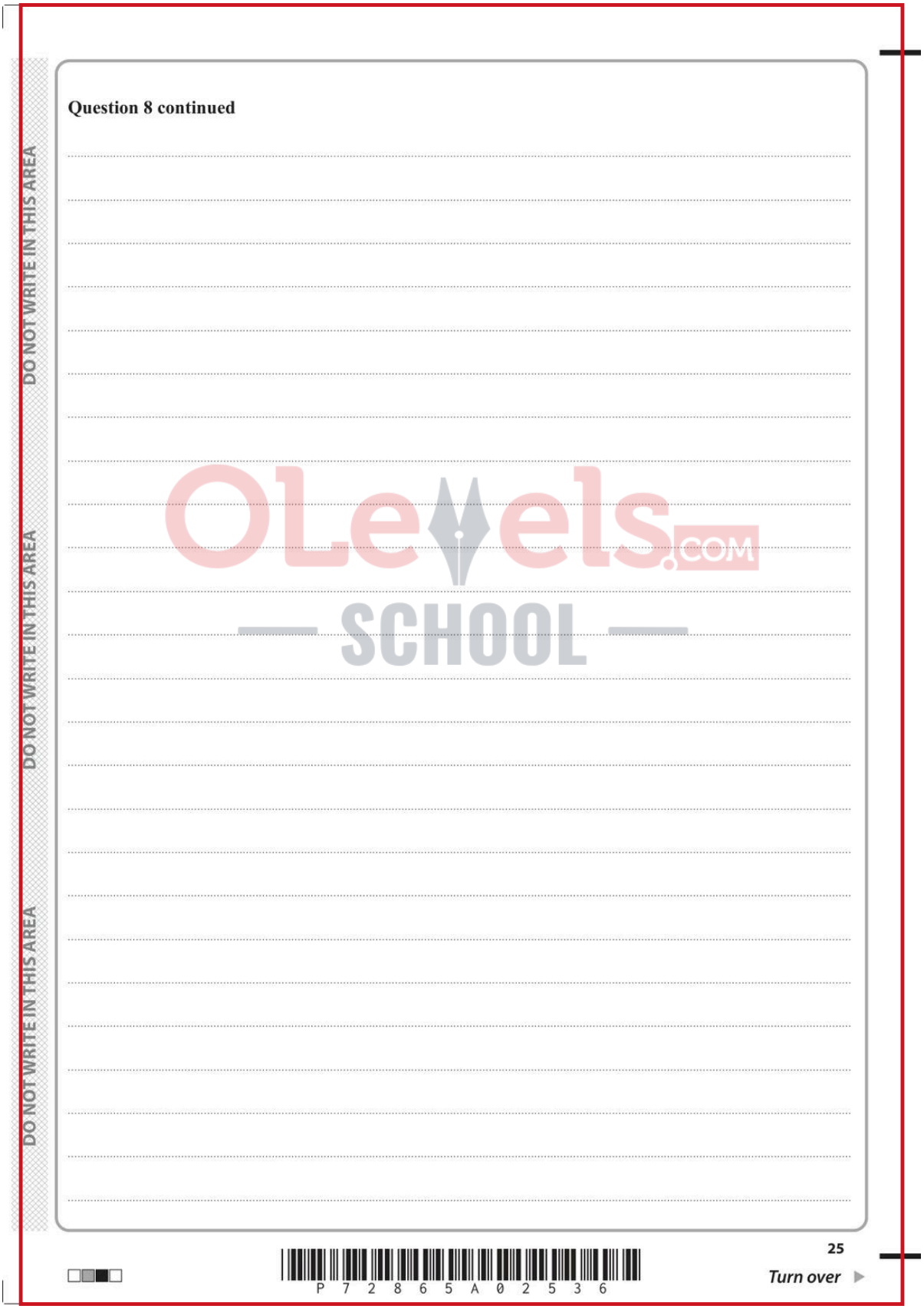
24

P 7 2 8 6 5 A 0 2 4 3 6

1. Question 8 · part e · 3 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part e only.



1. Question 8 · part e · 3 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part e only.

Question 8 continued

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1. Question 8 · part e · 3 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part e only.

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Question 8 continued

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(Total for Question 8 is 13 marks)

1. Question 8 · part e · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 8 · part d · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part d only.

8

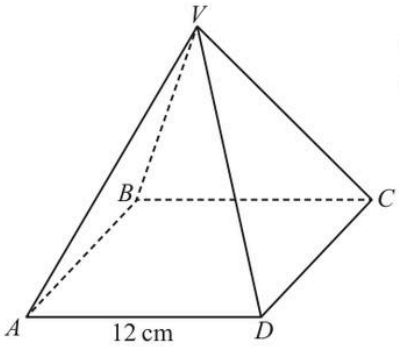


Diagram **NOT** accurately drawn

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(2)

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(3)

(c) Find, in cm^2, the exact area of triangle VAD

(3)

(d) Find, in cm, the exact length of the perpendicular from D to VA

(2)

(e) Find, in degrees to one decimal place, the size of the obtuse angle between the plane VAB and the plane VAD

(3)

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2. Question 8 · part d · 2 marks · Working and answer

Working and answer

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3. Question 11 · part c · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

11

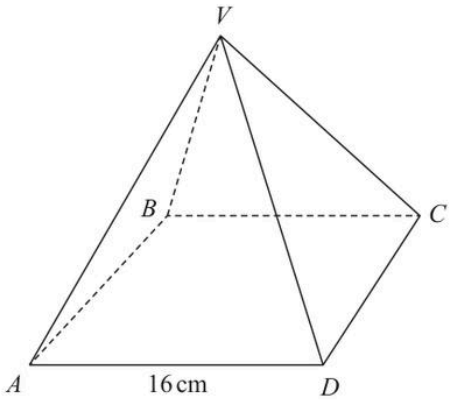


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right pyramid with vertex V and square base, $ABCD$, of side 16 cm.

The size of angle AVC is 90°

(a) Show that the height of the pyramid is $8\sqrt{2}$ cm.

(4)

(b) Find, in cm, the length of VA.

(3)

(c) Find, in cm, the exact length of the perpendicular from D onto VA.

(3)

Find, in degrees to one decimal place, the size of

(d) the angle between the plane VAB and the base $ABCD$,

(3)

(e) the obtuse angle between the plane VAB and the plane VAD.

(3)

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3. Question 11 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 11 · part e · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part e only.

11

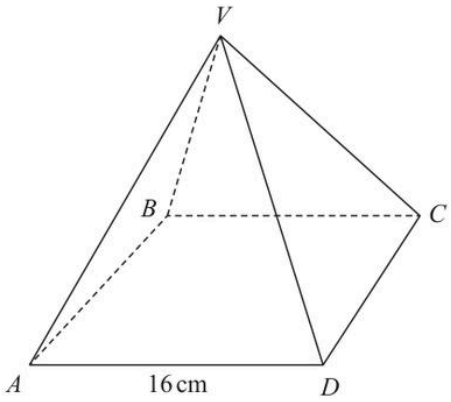


Diagram **NOT** accurately drawn

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(4)

(b) Find, in cm, the length of VA.

(3)

(c) Find, in cm, the exact length of the perpendicular from D onto VA.

(3)

Find, in degrees to one decimal place, the size of

(d) the angle between the plane VAB and the base $ABCD$,

(3)

(e) the obtuse angle between the plane VAB and the plane VAD.

(3)

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30

P 5 8 3 7 3 A 0 3 0 3 2

4. Question 11 · part e · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part e only.

Question 11 continued

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4. Question 11 · part e · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part e only.

Question 11 continued

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(Total for Question 11 is 16 marks)

4. Question 11 · part e · 3 marks · Working and answer

Working and answer

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5. Question 4 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

4

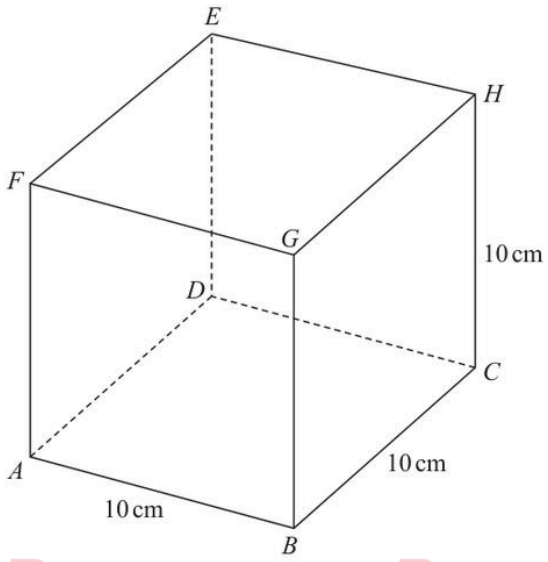


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a solid cube $ABCDEFGH$ with sides of length 10 cm.

(a) Find, in cm, the exact length of FC

(2)

(b) Find, in degrees to one decimal place, the size of $\angle FCA$

(2)

(c) Find, to the nearest degree, the size of the obtuse angle between the plane BFH and the plane BHC

(5)

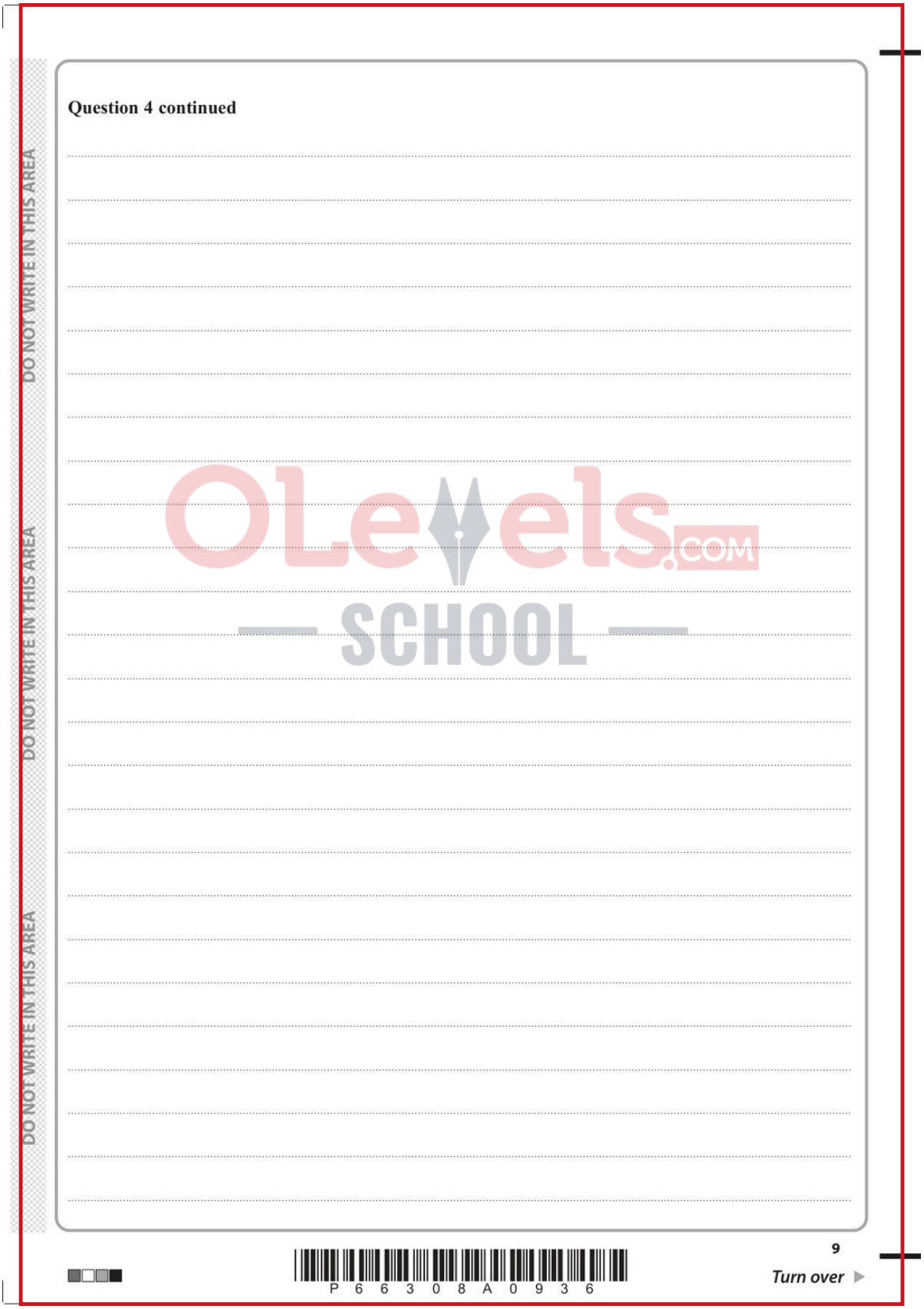
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5. Question 4 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.



5. Question 4 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

Question 4 continued

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5. Question 4 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 9 marks)

5. Question 4 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 10 · part d · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

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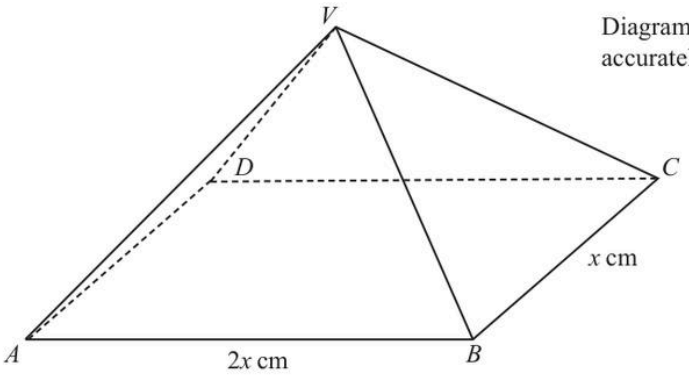


Figure 4

Figure 4 shows a right pyramid $ABCDV$

The base of the pyramid is a rectangle where,

$AB = DC = 2x$ cm $AD = BC = x$ cm

The edges VA, VB, VC and VD are all of equal length.

The angle between VA and $ABCD$ is 45°

(a) Show that $VA = \frac{\sqrt{10}}{2}x$ cm

(3)

(b) Find in cm, the exact height of the pyramid in terms of x

(2)

Find, in degrees to one decimal place,

(c) the size of angle VBA

(2)

(d) the size of the obtuse angle between the plane AVC and the plane BVD

(4)

Given that the volume of the pyramid is $9\sqrt{5}$ cm^3

(e) find the value of x

(2)

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P 7 4 2 8 5 R A 0 2 6 3 2

6. Question 10 · part d · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 11 · part c · 6 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

11

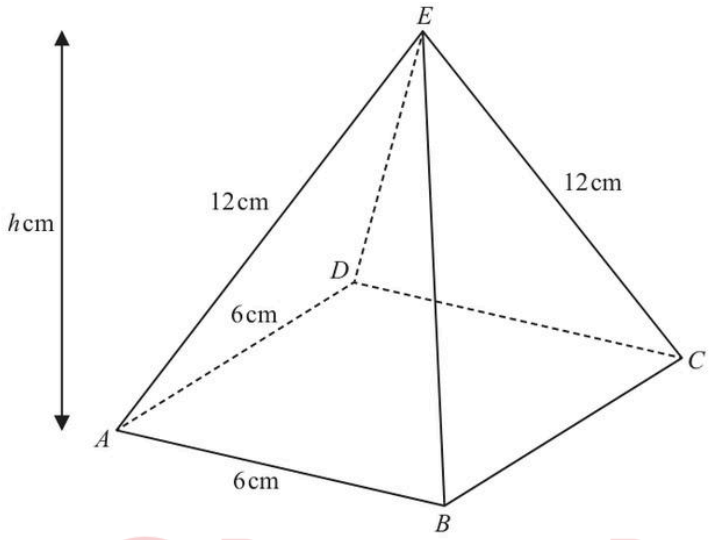


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a right pyramid $ABCDE$ with

- vertex E
- horizontal square base $ABCD$ of side 6 cm
- $EA = EB = EC = ED = 12$ cm
- vertical height of h cm

(a) Show that $h = 3\sqrt{14}$

(3)

(b) Find, in degrees to one decimal place, the size of the angle between EB and the plane $ABCD$

(2)

The midpoint of AE is P and the midpoint of BE is Q

(c) Find, to the nearest degree, the size of the obtuse angle between the plane EPQ and the plane $PQCD$

(6)

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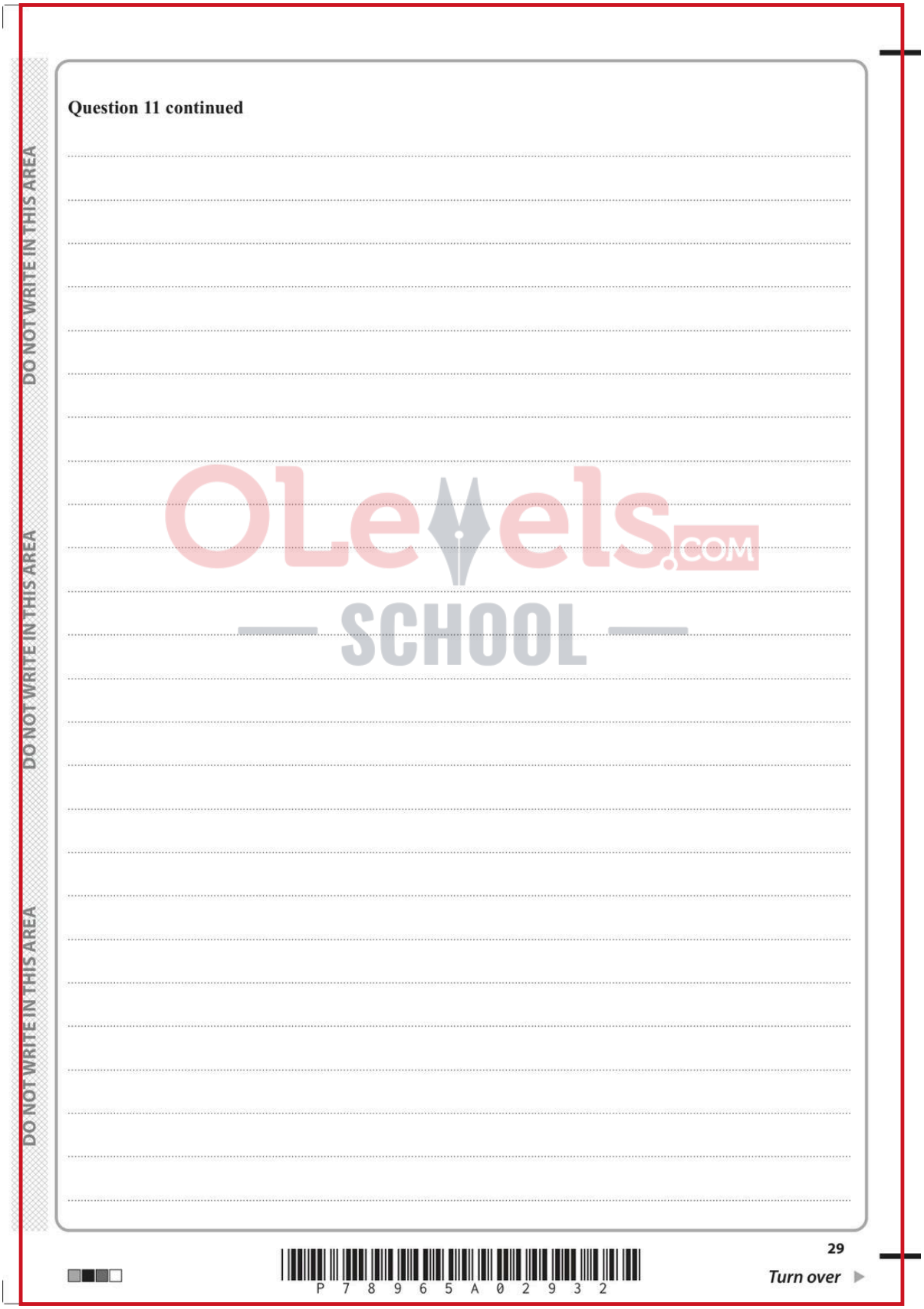
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P 7 8 9 6 5 A 0 2 8 3 2

7. Question 11 · part c · 6 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.



7. Question 11 · part c · 6 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

Question 11 continued

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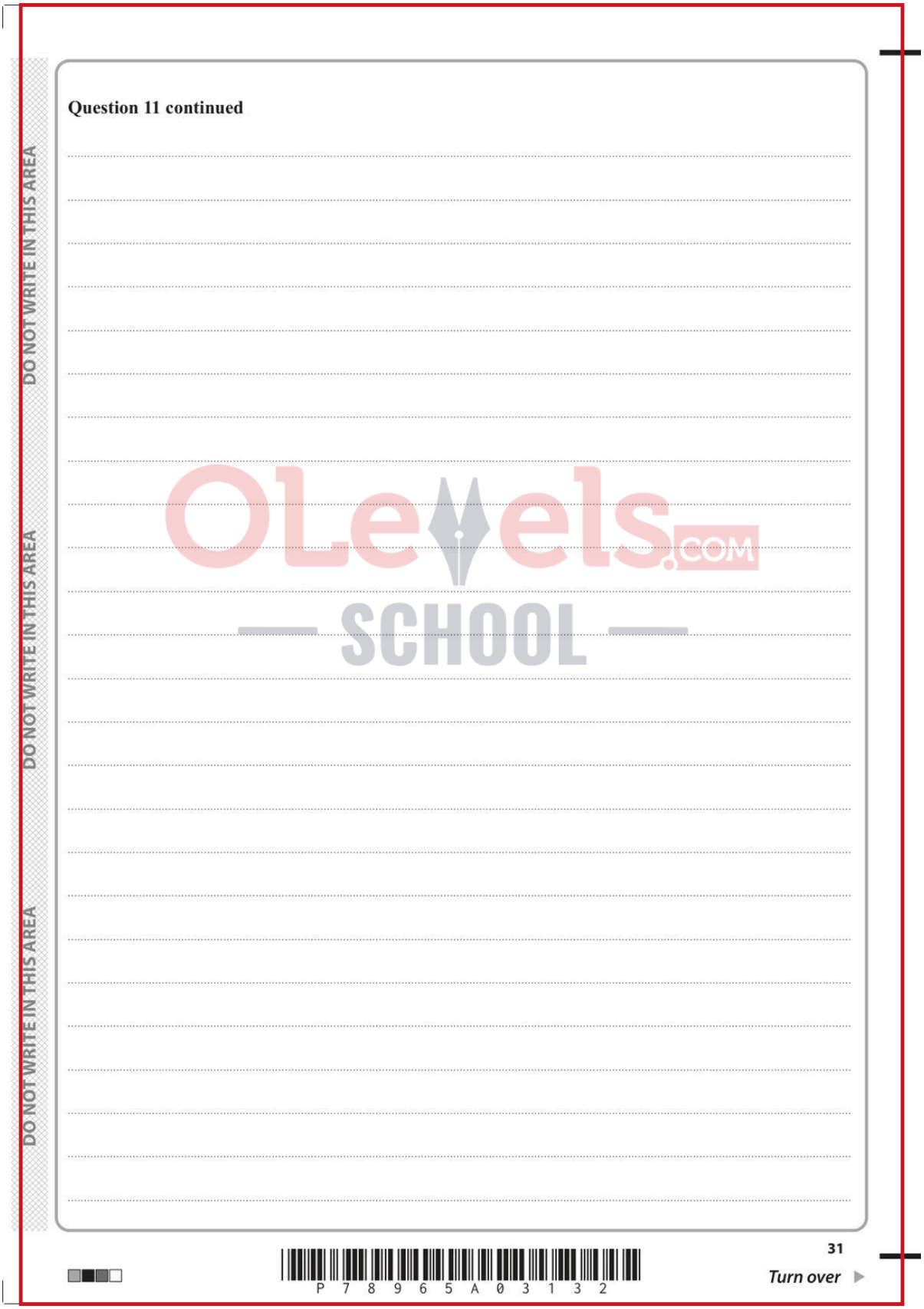
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7. Question 11 · part c · 6 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.



7. Question 11 · part c · 6 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

Question 11 continued

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(Total for Question 11 is 11 marks)

7. Question 11 · part c · 6 marks · Working and answer

Working and answer

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8. Question 9 · part c · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

9

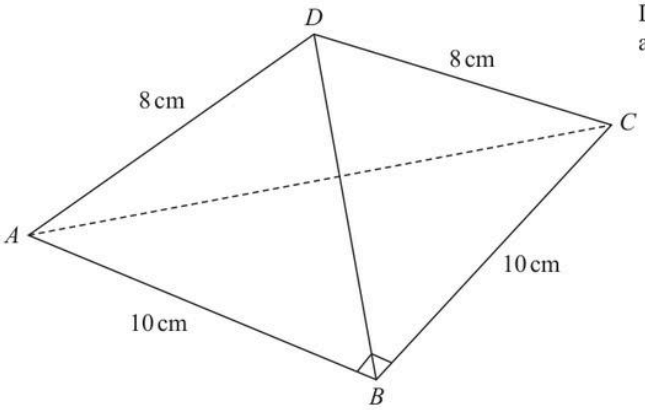


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a triangular pyramid $ABCD$ with base ABC

$AB = BC = 10$ cm $\quad AD = CD = 8$ cm $\quad \angle ABC = 90°$

(a) Find the exact length of AC

Give your answer in the form $p\sqrt{q}$ cm where p is an integer and q is a prime number.

(2)

The point M is the midpoint of AC

(b) Find the exact length of BM

Give your answer in the form $m\sqrt{n}$ cm where both m and n are prime numbers.

(2)

Given that $BD = 6$ cm,

(c) find, in degrees to one decimal place, the size of the acute angle between the plane ACD and the plane ABC

(4)

The base ABC of the pyramid is placed on a horizontal plane.

(d) Find, in cm to 3 significant figures, the vertical height of D above the base.

(2)

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8. Question 9 · part c · 4 marks · Working and answer

Working and answer



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9. Question 9 · part d · 4 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

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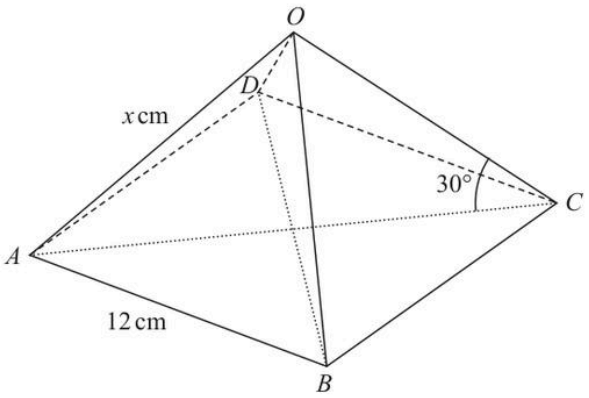


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows the right pyramid $OABCD$ with a square base $ABCD$ of side 12 cm.

$OA = OB = OC = OD = x$ cm and $\angle OAC = \angle ODB = \angle OCA = \angle OBD = 30°$

(a) Find the exact length of AC

(2)

(b) Show that $x = 4\sqrt{6}$

(2)

(c) Find the total surface area, to the nearest cm^2, of the pyramid.

(5)

(d) Find the size of the obtuse angle, to the nearest degree, between the plane OAB and the plane OBC

(4)

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9. Question 9 · part d · 4 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 9 continued

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9. Question 9 · part d · 4 marks

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Further practice · Attempt part d only.

Question 9 continued

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9. Question 9 · part d · 4 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 9 continued

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(Total for Question 9 is 13 marks)

9. Question 9 · part d · 4 marks · Working and answer

Working and answer

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